D1.3 Mutation and gene editing

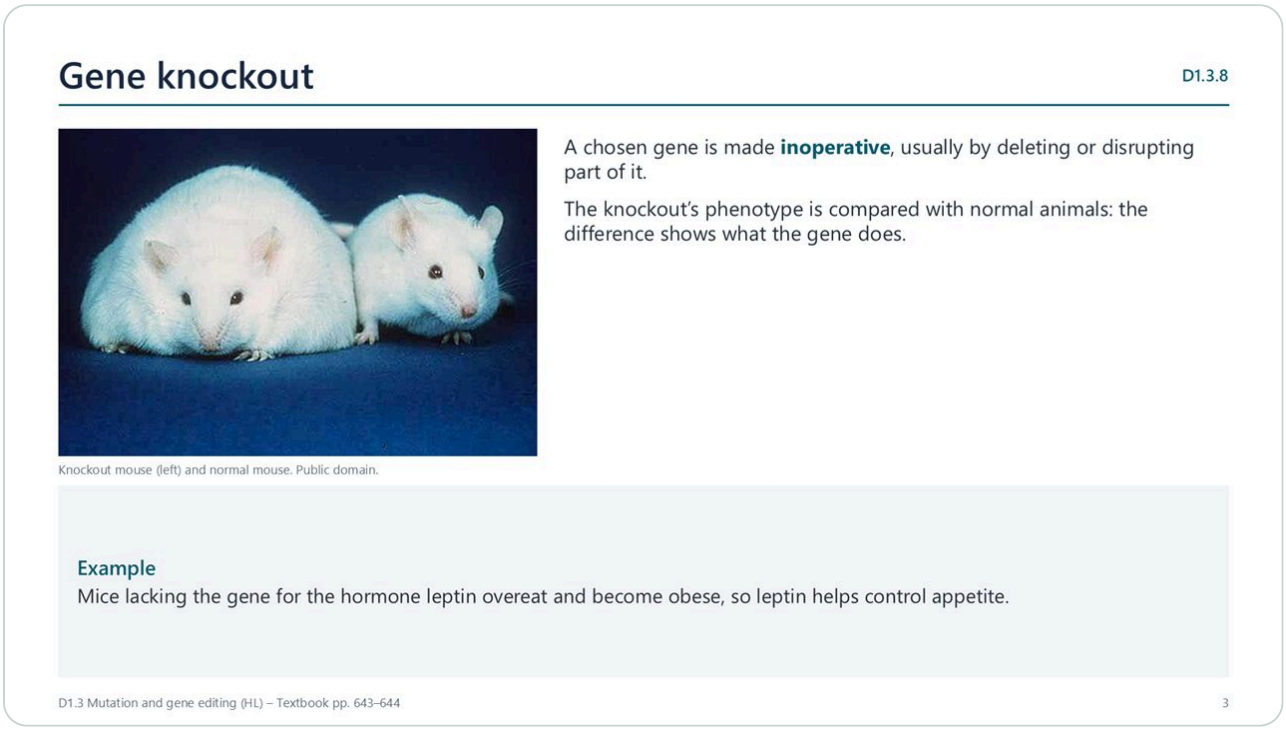
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d1-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 643–647 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

STUDYING AND EDITING GENES

D1.3.8 Gene knockout

pp. 643–645



A **gene knockout** makes a chosen gene inoperative, usually by deleting or disrupting part of it. Comparing the knockout organism with normal ones shows what the gene does: mice without p53 develop many tumours; mice without the leptin gene overeat and become obese. Libraries of knockout strains exist for model species such as mice. Results may not transfer exactly to humans, and other genes can compensate.

D1.3.9 CRISPR-Cas9

pp. 645–646

Scientists design a **guide RNA** whose bases match the target DNA. It binds the enzyme **Cas9**, which is led to the matching sequence and cuts both strands. When the cell repairs the cut, the gene can be disabled or new DNA inserted. A successful use: blood stem cells from people with sickle cell disease are edited so their red blood cells make foetal haemoglobin, which does not sickle; the treatment was approved in 2023.

D1.3.10 Conserved sequences

Conserved sequences are identical or similar across a species or group; **highly conserved** sequences have stayed similar over long periods of evolution, like histone genes in peas and humans. Two hypotheses explain this: the product only works with that sequence, so **selection removes changes**; or those regions have a **slower rate of mutation**.

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